Representational Measurement of Similarity: The Additive Difference Model, Revisited

Arthur Liske, Mathematik

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Abstract text......

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5.1 Preliminary lemma

Nur kurze zusammenfassung der Notation zur Übersicht Notation:

- γ: parametric curve, $\mathrm{L}(\gamma)$: length of curve, $\mathrm{L}_a^x(\gamma)$: length of subcurve from a to x
- δ_∞: ∞-metric, if no subscript then δ is the additive difference metric
- δ: metric, $\delta\,(a,\ b)$ metric distance between a and b, $\delta\,(a) := \delta\,(0,\ a)$
- $\mathrm{B}_\delta\,(r) := \{x \in \mathbb{R}^n : \delta\,(x) < r\}$: the open Ball with radius r around 0 wrt. the metric δ,
- $\mathrm{B}_\delta\,[r] := \{x \in \mathbb{R}^n : \delta\,(x) \leq r\}$: the closed Ball with radius r around 0 wrt. the metric δ,
- $\|x, y\|_\infty := \max_{i \leq n}\{|x_i - y_i|\}$: the maximum metric. We just use ∞ to shorten the notation, e.g. in $\mathrm{B}_\infty\,(r)$.
- $\mathrm{conv}(A)$: convex Hull of A.
- $\mathrm{extreme}(A)$: extreme Points of A.

In this section we will prove the main lemma, which is needed for the proof of the main result First, we will state the lemma, then give some remarks on the lemma and prove one statement needed for the lemma and then finally prove it. The proof is divided into 5 parts and a whole subsection which is devoted to showing a statement that is needed in the proof. Everything in this section will be assumed as in this Theorem. For example, if we write δ, this implies not any metric but the metric from the lemma with its properties. Moreover all topological terms(open, closed, continuous) are meant with respect to the natural topology of $\mathbb{R}^n$ (i.e. the topology induced by the δ_∞ and the euclidean metric), not the topology induced by δ or the induced topology on S. This will be especially important for the subsection on convexity of the balls. We will begin with the statement:

Lemma 5.7 (Main Lemma)**.**
Let S be an open convex subset of n-dimensional Euclidean space. Let δ be a metric on S that satisfies intradimensional subtractivity [1] with respect to a continuous function $F : \mathbb{R}^n_{\geq 0} \to \mathbb{R}_{\geq 0}$ and linear coordinates [2] in S. If $\langle S, \delta\rangle$ is a metric with additive segments, then δ is homogeneous, i.e., for any $x, z \in S$ and any real $0 \leq t \leq 1$,

$$\delta[x, x + t(z - x)] = t\,\delta(x, z).$$

Now we will continue with a remark that makes it possible to move the set such that $0 \in S$ by using the translation invariance of δ.

[1] i.e. δ takes the form $\delta\,(x,\ y) = F(|x_1 - y_1|, \ldots, |x_n - y_n|)$.

[2] Linear coordinates means that the vector representations are cartesian as opposed to, for example, polar, where each coordinate entry is not the value on the corresponding factor of the cartesian product.

Remark 5.8 (Translation invariance and domain of δ and F). (i) δ is translation-invariant, i.e. $\delta\,(x+y,\ z+y) = \delta\,(x,\ z)$:

$$\begin{aligned}\delta\,(x+y,\ z+y) &= F[|x_1+y_1-(z_1+y_1)|,\dots,x_n+|y_n-(z_n+y_n)|]\\ &= F[|x_1-z_1|,\dots,|x_n-z_n|]\\ &= \delta\,(x,\ z)\end{aligned}$$

(ii) Because δ is invariant to translation in its domain, we can move S and redefine the metric on the shifted set. We do this by defining $S' = \{s-x : s \in S\}$ for $x \in S$ and $\delta'(a,b) := \delta\,(a+x,\ b+x)$ for $a,b \in S'$. Since $0 \in S'$, we also can introduce the following shorthand Notation $\delta'(x') := \delta'(0,x')$ for $x' \in S'$. From now on, we will assume that $0 \in S$, because by the argument just provided, we can shift S and δ such that $0 \in S$. Thus we can proceed to proof the statement using the shorthand notation, by which we need to show that:

$$\delta\,(t\cdot z) = t\cdot\delta\,(z) \qquad \forall t \in [0,1] \text{ and } \forall z \in S'.$$

(iii) Since δ is defined by $\delta\,(x,\ y) = F(|x_1-y_1|,\dots,|x_n-y_n|)$ and δ is a metric, $F(x) = 0$ iff $x = 0$ must hold. Moreover the function F has to be defined on a set, where at least the "greatest possible difference" in coordinate i is a possible i-th argument. This means F is defined on a cartesian product of open, real intervals of the form $[0,a_i)$ where

$$a_i = \sup\{|x_i-y_i| : (x_1,\dots,x_n),(y_1,\dots,y_n) \in S\} \in \mathbb{R}_{>0}\cup\{\infty\}.$$

These a_i are these "greatest possible differences" on the i-th coordinate.

(iv) Moreover, due to the continuity of F and $|\cdot|$, the metric δ is continuous.

Before moving to the proof, we will fist show two statements that we will need right at the start. The first one asserts that the distance between a compact and a closed set is greater zero.

Todo: ref?

Lemma 5.9.
If $I \subset \mathbb{R}^n$ is compact, $Y \subset \mathbb{R}^n$ is closed and I and Y are disjoint, then there exists $\varepsilon > 0$ such that $\delta_\infty\,(p,\ y) > \varepsilon$ for all $p \in I$ and $y \in Y$.

Proof. We will prove this statement by contradiction. Assume that for all $\varepsilon > 0$ there exists $p \in I$ and $y \in Y$ such that $\delta_\infty\,(p,\ y) < \varepsilon$. This means, we can construct two sequences $p_n \in I$ and $y_n \in Y$ such that $\delta_\infty\,(p_n,\ y_n) \to 0$ as $n \to \infty$. I is compact, thus there exists a convergent subsequence p_{n_m} such that $p_{n_m} \to p \in I$ for $n \to \infty$. With the triangle inequality we get

$$\delta_\infty\,(p,\ y_{n_m}) \le \delta_\infty\,(p,\ p_{n_m}) + \delta_\infty\,(p_{n_m},\ y_{n_m}) \to 0 \text{ as } m \to \infty.$$

Hence, p is a limit point of Y, and since Y is closed, $p \in Y$ must hold which contradicts the assumption. $\square$

The second one allows us to make arbitrary small δ-balls which we will combine with the statement before to assure a distance in δ and not δ_∞.

Lemma 5.10 (arbitrary small δ-Balls)**.**
Let $\varepsilon > 0$ such that $\mathrm{B}_\infty(\varepsilon) \subset S$. Then there exists $\alpha > 0$ such that $\mathrm{B}_\delta(\alpha) \subset \mathrm{B}_\infty(\varepsilon)$.

Proof. Since δ is a metric defined by $\delta(x, y) = F[|x_1 - y_1|, \dots, |x_n - y_n|]$, we can define the j-th partial function of F by setting all other arguments to zero, i.e. $F^j(u_j) := F(0, \dots, u_j, \dots, 0)$.

F is strictly increasing in every argument implies that F^j is increasing and the continuity of F directly translates to F^j being continuous. Moreover, due to remark 5.8 (iii), $F^j(u_j) = 0$ if and only if $u_j = 0$ holds.

Together, this means that F^j is an increasing, continuous real function with $F^j(0) = 0$.

By setting $\alpha_j = F^j(\varepsilon)$ we get by monotonicity, that $F^j(u_j) < \alpha_j \implies u_j < \varepsilon$. Note, that $F^j(\varepsilon)$ is defined, because $\mathrm{B}_\infty(\varepsilon) \subset S'$ implies that ε is a possible distance for each coordinate. (See also remark 5.8 (iii). Using the a_i from the remark, we get that $\varepsilon < a_i$ holds for all i and thus $F^j(\varepsilon)$ is defined for all $j = 1, \dots n$.)

Doing this for $j = 1, \dots, n$ yields $\alpha_1, \dots, \alpha_n$. By setting $\alpha = \min\{\alpha_1, \dots, \alpha_n\}$ the above holds for every partial function F^j which yields the boundary for the ∞ Metric:

$$F^j(u_j) < \alpha \leq \alpha_j \implies u_j < \varepsilon \quad \forall j = 1, \dots, n \implies \|u\|_\infty < \varepsilon.$$

Using the monotonicity of F we get

$$\delta(u) = F(u) < \alpha \implies F^j(u_j) < \alpha \implies \|u\|_\infty < \varepsilon.$$

□

Now, we will move to the proof of the main lemma. It is suggested to first read the proof to get the main ideas and a big picture understanding, and then read the next section which covers the convexity of spheres. First we will restate the lemma in the form, in which we will prove it.

Lemma.
Let S be an open convex subset of n-dimensional Euclidean space with $0 \in S$. Let δ be a metric on S that satisfies intradimensional subtractivity with respect to linear coordinates in S. If $\langle S, \delta \rangle$ is a metric with additive segments, then δ is homogeneous, i.e., for any $z \in S$ and any real $0 \leq t \leq 1$,

$$\delta(tz) = t\,\delta(z).$$

The proof consists of 4 parts. First we will find a radius such that all balls with this radius on line from 0 to z are in S. Then we will prove this lemma for $t = \frac{1}{m}$ for $m \in \mathbb{N}$ by proving the two inequalities. After this, we will expand this statement to the rationals and then use continuity to expand it to all real numbers between 0 and 1.

Proof.

Part 0 (Asserting a common distance for all points on the line segment)**.**
Let $z \in S$. Since S is open, $\mathbb{R}^n \setminus S$ is closed. The set $I := \{tz : t \in [0,1]\}$ is compact, since continuous images($t \mapsto tz$) of compact sets ($t \in [0,1]$) are compactTodo: ref? . Thus, by lemma 5.9 there exists a distance $\varepsilon > 0$ such that $\delta_\infty(p,\ y) < \varepsilon$ for all $p \in I$ and $y \in \mathbb{R}^n \setminus S$. This means, for all points on the line from 0 to z the ball with radius ε is a subset of S, i.e. for all $t \in [0,1] : \mathrm{B}_\infty(tz,\ \varepsilon) \subset S$. By lemma 5.10, there exists $\alpha > 0$ such that $\mathrm{B}_\delta(\alpha) \subset \mathrm{B}_\infty(\alpha)$, and by translation invariance this holds for all points on the line: $\mathrm{B}_\delta(\alpha)\, tz \subset S$ for all $t \in [0,1]$. Now let $m \in \mathbb{N}$ such that $\frac{1}{m}\,\delta(z) < \alpha$. In particular, this means that $\mathrm{B}_\delta(\alpha)$ and therefore, $\mathrm{B}_\delta\left(\frac{1}{m}\,\delta(z)\right)$ is bounded (with respect to the ∞ metric).

Part 1 ($t\,\delta(z) \leq \delta(tz)$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$)**.**
Let $z^{(k)} := \frac{k}{m} \cdot z$ for $k = 0, \ldots, m$. $z^{(k)} \in S$ for all k due to convexity of S.

By applying translation invariance, we can calculate the distance between two consecutive points:

$$\delta\left(z^{(k-1)},\ z^{(k)}\right) = \delta\left(\frac{k-1}{m}z,\ \frac{k-1}{m}z + \frac{1}{m}z\right) = \delta\left(\frac{1}{m}z\right) \quad \forall k = 1, \ldots, m.$$

Applying the triangle inequality $m-1$-times and using the above result yields

$$\delta(z) \leq \sum_{k=0}^{m-1} \delta\left(z^{(k)},\ z^{(k+1)}\right) = m \cdot \delta\left(\frac{1}{m}z\right).$$

Finally, dividing the right side by m yields $\frac{1}{m}\,\delta(z) \leq \delta\left(\frac{1}{m}z\right)$, which concludes the first part.

Part 2 ($t\,\delta(z) \geq \delta(tz)$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$)**.**
Since δ is a metric with additive segments, there exists a segment from 0 to z with length $\delta(z)$. **??** yields, that the arc length is a continuous, non-decreasing function. This means, we can divide the segment into m subsegments with length $\frac{1}{m}\,\delta(z)$ and choose $y^{(0)}, \ldots, y^{(m)}$ equally spaced on the segment such that $\delta\left(y^{(i)},\ y^{(i-1)}\right) = \frac{1}{m}\,\delta(z)$ and $y^{(0)} = 0$ and $y^{(m)} = z$. Using this, we can define $x^{(i)} := y^{(i)} - y^{(i-1)}$. Translation invariance yields $\delta\left(x^{(i)}\right) = \delta\left(y^{(i)},\ y^{(i-1)}\right) = \frac{1}{m}\,\delta(z)$ and, due to the choice of m in part 0 $x^{(i)} \in S$ holds. Note, that this is still true, if we increase m.

Further, we can observe that $\frac{1}{m}z$ is a convex combination of $x^{(i)}$:

$$\sum_{i=1}^{m} \frac{1}{m} x^{(i)} = \sum_{i=1}^{m} \frac{1}{m}(y^{(1)} - y^{(0)} + y^{(2)} - y^{(1)} + \cdots + y^{(m)}) = \frac{1}{m}(-y^{(0)} + y^{(m)}) = \frac{1}{m}z$$

We showed before, that $\delta\left(x^{(i)}\right) = \frac{1}{m}\,\delta(z)$, which implies $x^{(i)} \in \mathrm{B}_\delta\left[\frac{1}{m}\,\delta(z)\right]$. By part 0 $\mathrm{B}_\delta\left(\frac{1}{m}\,\delta(z)\right)$ is bounded. Therefore we can apply lemma 5.11 from which we get that spheres are convex. This means, that the spheres are closed under convex combination, which implies that $\frac{1}{m}z \in \mathrm{B}_\delta\left[\frac{1}{m}\,\delta(z)\right]$ must hold. Thus $\delta\left(\frac{1}{m}z\right) \leq \frac{1}{m}\,\delta(z)$ which concludes the second part of the proof.

Now, by combining part 1 and part 2, we have established the equality for $t = \frac{1}{m}$. Note, that the equality also holds for all $\mathbb{N} \ni m' > m$ since we only needed m to be big enough in part 2 for $x^{(i)}$ to be in S and for convexity of the spheres. Thus, by making m bigger, the crucial points in part 2 and therefore the equality are still true. Next, we will proceed to extending the equality from $t = \frac{1}{m}$ to $t = \frac{k}{m}$ for $k = 0, \ldots, m$.

Part 3 ($t\,\delta(z) = \delta(tz)$ for $t = \frac{k}{m}$ and $t \in \{0, \ldots, m\}$).
For $z^{(1)}$ we already have $\delta\left(0,\ z^{(1)}\right) = \frac{1}{m}\,\delta(0,\ z)$. Now, for consecutive points, by translation invariance, it holds that

$$\delta\left(z^{(k-1)},\ z^{(k)}\right) = \delta\left(z^{(k-2)},\ z^{(k-1)}\right) = \cdots = \delta\left(z^{(1)},\ z^{(0)}\right) = \delta\left(z^{(1)},\ 0\right) = \frac{1}{m}\,\delta(0,\ z).$$

Using this and the triangle inequality, we get

$$\begin{aligned}
\delta(0,\ z) &\leq \delta\left(0,\ z^{(1)}\right) + \delta\left(z^{(1)},\ z^{(2)}\right) + \cdots + \delta\left(z^{(m-1)},\ z^{(m)}\right) \\
&= n \cdot \frac{1}{n}\,\delta(0,\ z) \\
&= \delta(0,\ z) \\
\Longrightarrow\ \delta(0,\ z) &= \delta\left(0,\ z^{(1)}\right) + \delta\left(z^{(1)},\ z^{(2)}\right) + \cdots + \delta\left(z^{(m-1)},\ z^{(m)}\right)
\end{aligned}$$

For the distance $\delta\left(0,\ z^{(2)}\right)$, we get the following two inequalities:

1. $$\delta\left(0,\ z^{(2)}\right) \leq \delta\left(0,\ z^{(1)}\right) + \delta\left(z^{(1)},\ z^{(2)}\right) = \frac{2}{m}\,\delta(0,\ z)$$

2. $$\begin{aligned}
\delta(0,\ z) &\leq \delta\left(0,\ z^{(2)}\right) + \delta\left(z^{(2)},\ z\right) \\
\delta\left(0,\ z^{(2)}\right) &\geq \delta(0,\ z) - \underbrace{\delta\left(z^{(2)},\ z\right)}_{\leq \sum_i \delta\left(z^{(2+i)},\ z^{(2+i+1)}\right)} \\
&\geq \delta(0,\ z) - \sum_{i=0}^{m-3} \delta\left(z^{(2+i)},\ z^{(2+i+1)}\right) \\
&= \delta(0,\ z) - \frac{m-2}{m}\,\delta\left(0,\ z^{(k)}\right) \\
&= \frac{2}{m}\,\delta(0,\ z)
\end{aligned}$$

These two inequalities yield $\delta\left(0,\ z^{(2)}\right) = \frac{2}{m}\,\delta(0,\ z)$. By repeating this for $\delta\left(0,\ z^{(k)}\right)$ for every $k = 2, \ldots, m-1$, we get the equality for every rational number for m big enough. However every rational number with a denominator smaller than m can be represented by another rational number with denominator greater than m. This yields, that the result is true for every rational t.

Part 4 ($t\,\delta(z) = \delta(tz)$ for $t \in [0,1]$)**.**
The final step can be completed using continuity of the metric. Now let $t \in [0,1]$ and $t_n \in \mathbb{Q}$ with $t_n \to t$ for $n \to \infty$. Then

$$\begin{aligned}
\delta(tz) &= \delta\left(\lim_{n\to\infty} t_n z\right) \\
&= \lim_{n\to\infty} \delta(t_n z) && \text{(continuity)} \\
&= \lim_{n\to\infty} t_n\,\delta(z) && (t_n \in \mathbb{Q} \text{ and part 3}) \\
&= t\,\delta(z)
\end{aligned}$$

$\square$

5.1.1 Convexity of δ Balls

In this subsection we will proof the following lemma which is needed to conclude the proof of lemma 5.7. Everything will be assumed as in the lemma 5.7. Note, that all topological terms are with respect to the natural topology of the euclidean space.

Lemma 5.11 (Convexity of δ-Spheres).
If $\mathrm{B}_\delta[\alpha] \subset S$ *is bounded, then it is convex.*

First, we will state some definitions and results on basic convex geometry that we need, and prove the theorem at the end of the section. The definitions and results on convex geometry are taken out of (Hug & Weil, 2020).

First of all, we need the notions of convex combination, the convex hull and extreme points. A convex combination is similar to a linear combination, but has the additional restraint that the coefficients must be positive ad add up to 1. The convex hull of a set is, like any other hulls, the smallest convex set that contains the set.

Definition 5.12 (Convex Combination and Hull).

1. Let $k \in \mathbb{N}$, let $x_1, \ldots, x_k \in \mathbb{R}^n$, and let $a_1, \ldots, a_k \in [0,1]$ be such that $a_1 + \cdots + a_k = 1$. Then $a_1 x_1 + \cdots + a_k x_k$ is called a *convex combination* of the points $x_1, \ldots x_k$.
2. For $A \subset \mathbb{R}^n$ the *convex Hull* $\mathrm{conv}(A)$ is the smallest convex set containing A, i.e.

$$\mathrm{conv}(A) := \cap \{C \subset \mathbb{R}^n : A \subset C, C \text{ convex}\}$$

The next theorem sums up the relationship between convex combination and convex hulls.

Theorem 5.13.
For $A \subset \mathbb{R}^n$ *the convex Hull is the set of all convex combination of points in* A*:*

$$\mathrm{conv}(A) = \left\{ \sum_{i=1}^{k} a_i x_i : k \in \mathbb{N}, x_1, \ldots, x_k \in A, a_1, \ldots, a_k \in [0,1] : \sum_{i=1}^{k} a_i = 1 \right\}$$

Moreover, the convex hull preserves compactness, as the following theorem shows.

Theorem 5.14 (convex Hull of compact set is compact).
If $A \subset \mathbb{R}^n$ *is compact, then* $\mathrm{conv}(A)$ *is compact.*

Next, we will introduce extreme points, which are, in a sense, characteristic points of a set that cannot be represented by a (strict) convex combination.

Definition 5.15 (Extreme Point).
Let $A \subset \mathbb{R}^n$ be closed and convex. A point $x \in A$ is called an *extreme point of* A if x cannot be represented as a nontrivial (or strict) convex combination of points of A, that is, if $x = a \cdot y + (1-a) \cdot z$ with $y, z \in A$ and $a \in (0,1)$, implies that $x = y = z$. The set of all extreme points of A is denoted by $\mathrm{extreme}(A)$.

For compact and convex sets, the extreme points are enough to "build" the convex hull of the set by doing convex combinations. In particular, this means that for a compact and convex set, every point can be expressed as a convex combination of extreme points.

Theorem 5.16 (Minkowski Theorem)**.**
Let $A \subset \mathbb{R}^n$ be compact and convex. Then $A = \operatorname{conv}(\operatorname{extreme} A)$.

Before moving to the convexity of balls, we will first prove two lemma, which we will need while proving convexity of spheres. The first one is the openness/closedness of δ-balls.

Lemma 5.17 (δ-Balls are open/closed)**.**
For every radius $r > 0$, the δ-Balls are open $(\mathrm{B}_\delta(r))$ or closed $(\mathrm{B}_\delta[r])$.

Proof. $\delta : S' \times S' \to \mathbb{R}_{\geq 0}$ is continuous with $S' \subseteq \mathbb{R}^n$ and therefore $y \mapsto \delta(y) = \delta(0,\ y)$ is continuous.

Consider the closed interval $U' = [0, \alpha] \subset \mathbb{R}_{\geq 0}$. Then

$$\begin{aligned}
\delta^{-1}(U') &= \{y \in S' : \delta(y) \in U'\} \\
&= \{y \in S' : \delta(0,\ y) \leq \alpha\} \\
&= \mathrm{B}_\delta[\alpha]
\end{aligned}$$

Due to topological continuity of δ (which is equivalent to preimages of closed sets being closed) $\mathrm{B}_\delta[\alpha]$ is closed. By translation invariance this holds for closed balls with any center in S. Todo: ref topo stetig? Moreover, we can obtain the same result for open balls by repeating this proof and replacing the closed interval with an open one and $\leq$ with $<$. $\square$

The second lemma shows, that extreme points of balls get compressed by a factor of $\alpha \in \mathbb{N}$, if the radius of the ball gets shrinked by the same factor.

Lemma 5.18 (compressing extreme points)**.**
Let $\alpha \in \mathbb{R}_{>0}$ and $r \in \mathbb{N}_{>0}$. Further let $\mathrm{E}(\alpha) := \operatorname{extreme}(\mathrm{B}_\delta[\alpha])$ and $\mathrm{C}[\alpha] := \operatorname{conv}(\mathrm{B}_\delta[\alpha])$. Then, $y \in \mathrm{E}(\alpha)$ implies $\frac{1}{r}y \in \mathrm{E}\left(\frac{\alpha}{r}\right)$.

Proof. We will prove this statement by contraposition.

Assume, that $\frac{1}{r}y$ is not an extreme point, i.e. $\frac{1}{r}y \in \mathrm{C}\left[\frac{\alpha}{r}\right] \setminus \mathrm{E}\left(\frac{\alpha}{r}\right)$. Then we can write $\frac{1}{r}y$ as a strict convex combination of elements $w^i \in \mathrm{C}\left[\frac{\alpha}{r}\right]$:

$$\frac{1}{r}y = \sum_{i=1}^{p} \lambda_i w^i \quad \text{with} \quad \sum_{i=1}^{p} \lambda_i = 1, \quad \lambda_i > 0,\ w^i \in \mathrm{C}\left[\frac{\alpha}{r}\right] \text{ and } p \in \mathbb{N}$$

But because $\mathrm{C}\left[\frac{\alpha}{r}\right] = \operatorname{conv}(\mathrm{B}_\delta\left[\frac{\alpha}{r}\right])$ holds, we can write any element from $\mathrm{C}\left[\frac{\alpha}{r}\right]$ as a convex combination of elements from $\mathrm{B}_\delta\left[\frac{\alpha}{r}\right]$ and therefore we can assume $w^{(i)} \in \mathrm{B}_\delta\left[\frac{\alpha}{r}\right]$.

By translation invariance and the triangle inequality we get for all i:

$$
\begin{aligned}
\delta\left(rw^{(i)}\right) &\leq \delta\left(0,\ w^{(i)}\right) + \delta\left(w^{(i)},\ rw^{(i)}\right) && \text{triangle inequality}\\
&= \delta\left(w^{(i)}\right) + \delta\left(0,\ (r-1)\cdot w^{(i)}\right) && \text{translation invariance}\\
&\leq \delta\left(w^{(i)}\right) + \delta\left(0,\ w^{(i)}\right) + \delta\left(w^{(i)},\ (r-1)\cdot w^{(i)}\right) && \text{triangle inequality}\\
&= 2\cdot\delta\left(w^{(i)}\right) + \delta\left(0,\ (r-2)\cdot w^{(i)}\right) && \text{translation invariance}\\
&\dots\\
&\leq r\cdot\delta\left(w^{(i)}\right)\\
&= r\cdot\frac{\alpha}{r} = \alpha\\
\implies rw^{(i)} &\in \mathrm{B}_\delta\left[\alpha\right]
\end{aligned}
$$

This yields, that $y = \sum_{i=1}^{p} \lambda_i r w^{(i)}$ is a strict convex combination (because $\lambda_i r \neq 0$) and therefore y is not in $\mathrm{E}\left(\alpha\right)$. □

Now we can prove the convexity of δ-balls.
Todo: ref heine-borel?

Proof of lemma 5.11. Let $\mathrm{E}\left(\alpha\right) := \text{extreme}(\mathrm{B}_\delta\left[\alpha\right])$ and $\mathrm{C}\left[\alpha\right] := \text{conv}(\mathrm{B}_\delta\left[\alpha\right])$. By lemma 5.17 $\mathrm{B}_\delta\left[\alpha\right]$ is closed and by assumption it is also bounded. Therefore, from the Heine-Borel Theorem, it follows that $\mathrm{B}_\delta\left[\alpha\right]$ is compact.

By definition and theorem 5.14 it follows, that $\mathrm{C}\left[\alpha\right]$ is compact (and therefore closed, which is the reason why we denote it with square brackets) and convex. Application of the Minkowski Theorem (theorem 5.16) yields:

$$\mathrm{C}\left[\alpha\right] = \text{conv}(\text{extreme}(\mathrm{C}\left[\alpha\right]))$$

Further, it holds that $\text{extreme}(\mathrm{C}\left[\alpha\right]) = \text{extreme}(\mathrm{B}_\delta\left[\alpha\right])$ (By doing convex combinations, no new extreme points can be introduced because these are the points, which cannot be represented by a convex combination). Combining this and the statement above yields:

$$\mathrm{B}_\delta\left[\alpha\right] \subset \mathrm{C}\left[\alpha\right] = \text{conv}(\text{extreme}(\mathrm{B}_\delta\left[\alpha\right])).$$

Now, if we prove that $\text{conv}(\text{extreme}(\mathrm{B}_\delta\left[\alpha\right])) \subset \mathrm{B}_\delta\left[\alpha\right]$, we get that $\mathrm{B}_\delta\left[\alpha\right] = \mathrm{C}\left[\alpha\right]$ which proves the convexity of $\mathrm{B}_\delta\left[\alpha\right]$. We simplify this statement before proving it.

$$\mathrm{conv}(\underbrace{\mathrm{extreme}(\mathrm{B}_\delta\left[\alpha\right])}_{\mathrm{E}(\alpha)}) \subset B$$

$$\Updownarrow$$

$$\sum_{i=1}^{p} \lambda_i y^{(i)} \in \mathrm{B}_\delta\left[\alpha\right], \quad p \in \mathbb{N}_{>0}, \begin{cases} y^{(i)} \in \mathrm{E}\left(\alpha\right) \\ \lambda_i \in \mathbb{R}_{\geq 0}, \sum_{i=1}^{p} \lambda_i = 1 \end{cases} \qquad \text{(Definition)}$$

$$\Uparrow$$

$$\sum_{i=1}^{p} \lambda_i y^{(i)} \in \mathrm{B}_\delta\left[\alpha\right], \quad p \in \mathbb{N}_{>0}, \begin{cases} y^{(i)} \in \mathrm{E}\left(\alpha\right) \\ \lambda_i \in \mathbb{Q}_{\geq 0}, \sum_{i=1}^{p} \lambda_i = 1 \end{cases} \qquad (B \text{ closed, } \mathbb{Q} \text{ dense in } \mathbb{R}\ ^3)$$

$$\Uparrow$$

$$\sum_{i=1}^{r} \frac{1}{r} y^{(i)} \in \mathrm{B}_\delta\left[\alpha\right], \quad r \in \mathbb{N}_{>0},\ y^{(i)} \in \mathrm{E}\left(\alpha\right) \qquad (r \text{ common denominator})$$

Thus, we will continue proving that for $r \in \mathbb{R}_{>0}$ and $y^{(i)} \in \mathrm{E}\left(\alpha\right)$ it holds that $\sum_{i=1}^{r} \frac{1}{r} y^{(i)} \in \mathrm{B}_\delta\left[\alpha\right]$.

For $y^{(i)} \in \mathrm{E}\left(\alpha\right)$ it holds by lemma 5.18 that $\frac{1}{r} y^{(i)} \in \mathrm{E}\left(\frac{\alpha}{r}\right) \subset \mathrm{B}_\delta\left[\frac{\alpha}{r}\right]$. This means, that for the metric distance of the convex combination the following holds:

$$\begin{aligned}
\delta\left(0,\ \sum_{i=1}^{r} \frac{1}{r} y^{(i)}\right) &\leq \delta\left(0,\ \frac{1}{r} y^{(1)}\right) + \delta\left(\frac{1}{r},\ \sum_{i=1}^{r} \frac{1}{r} y^{(i)}\right) && \text{(triangle inequality)} \\
&= \delta\left(0,\ \frac{1}{r} y^{(1)}\right) + \delta\left(0,\ \sum_{i=2}^{r} \frac{1}{r} y^{(i)}\right) && \text{(translation invariance)} \\
&\leq \frac{\alpha}{r} + \delta\left(0,\ \sum_{i=2}^{r} \frac{1}{r} y^{(i)}\right) && \text{(translation invariance)} \\
&\dots && \text{(repeat)} \\
&\leq r \frac{\alpha}{r} = \alpha
\end{aligned}$$

This yields that $\sum_{i=1}^{r} \frac{1}{r} y^{(i)} \in \mathrm{B}_\delta\left[\alpha\right]$ holds which proves the statement. $\square$

[3] Density implies, that we can express every real number as a limit of rational numbers. B is closed, and therefore all limits of rational coefficients are included, if the statement holds for rational numbers.

Theorem 5.19 (Eindeutigkeitssatz zu l^p Metriken)**.**
Suppose $\langle A, \succsim \rangle$ is a complete, segmentally additive, factorial proximity structure, that is also and additive difference structure.
Then there exists a unique $r \geq 1$ and real-valued functions φ_i, defined on A_i, $i = 1, \dots, n$, that are unique up to a positive linear transformation, such that, for all $a, b, c, d \in A$,

$$(a, b) \succsim (c, d) \iff \delta(a, b) \geq \delta(c, d),$$

where

$$\delta(a, b) = \left[\sum_{i=1}^{n} |\varphi_i(a) - \varphi_i(b)|^r \right]^{1/r}.$$

Proof. Der Beweis passiert in mehreren Schritten. Zuerst sagen wir etwas über die Definitions- und Zielbereiche der Funktionen F_i und finden damit Grenzen für den Definitionsbereich von F. Danach argumentieren wir, dass wir F bis auf die positiven reellen Zahlen $[0, \infty)$ erweitern können und dass diese Erweiterung eine Funktionsgleichung impliziert, die nur von $f(x) = kx^p$ erfüllt wird. Im letzten Schritt zeigen wir, dass F auf ganz $\mathbb{R}^+$ gleich der Erweiterung sein muss und damit δ eine l^p Metrik sein muss.

Todo: Übersicht vom Beweis aufschreiben

$$F^{-1} \left[\sum_{i=1}^{n} F(t\alpha_i) \right] = tF^{-1} \left[\sum_{i=1}^{n} F(\alpha_i) \right] \tag{2}$$

Todo: necessary sufficient stuff benutzen und zeigen, dass wir das Lemma nutzen können!

Part 1.
By theorem 5.1, the additive difference model

$$\delta(a, b) = G\left(\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right) \right)$$

yields a metric (that is additive on the coordinate axes) if $F_i(\alpha) = F(t_i\alpha)$ and $F = G^{-1}$ for $i = 1, ..., n$. Thus, F^{-1} is defined for all numbers $\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)$, and each F_i coincides with F on its domain (except for the factor t_i, which scales the argument and adjusts the domain).
Since the domains of F_i are intervals (because δ is a metric with additive segments) Todo: begründung checken , we can choose $\omega > 0$ such that for each $i = 1, \dots, n$ and any α with $0 \leq \alpha \leq \omega/t_i$, α lies within the domain of F_i.
In particular, if we set $\omega_i = \omega/t_i$, then $F^{-1}\left[\sum_{i=1}^{n} F_i(\omega_i)\right] = F^{-1}\left[nF(\omega)\right] =: \Omega$ is defined. Thus, $F = G^{-1}$ is defined at least on the interval $[0, \Omega]$. We can now extend F as follows.

Part 2.
For $\alpha < \omega$, we have by eq. (2),

$$\begin{aligned} F^{-1}[nF(\alpha)] &= F^{-1}[nF(\alpha\omega/\omega)] \\ &= (\alpha/\omega)F^{-1}[nF(\omega)] \\ &= \alpha\Omega/\omega. \end{aligned}$$

Thus F satisfies the following, by plugging in the correct values for α ($\tilde{\alpha} = \alpha\frac{\omega}{\Omega}$ and $\tilde{\alpha} = F^{-1}(\frac{\alpha}{n})$)

$$F(a) = nF(a\omega/\Omega), \quad 0 \leq a \leq \Omega; \tag{3}$$

$$F^{-1}(a) = (\Omega/\omega)F^{-1}(a/n), \quad 0 \leq a \leq nF(\omega). \tag{4}$$

These two equations can now be used to extend the domain of F and F^{-1}.
We use eq. (3) by defining F through the right-hand side (for values where the right-hand side is defined, i.e., for $0 \leq a \leq \Omega^2/\omega$). Since $\Omega^2/\omega > \Omega$, this indeed extends F beyond the interval $[0, \Omega]$.
Similarly, F^{-1} satisfies equation eq. (4) for the extended interval $0 \leq a \leq nF(\Omega) = n^2F(\omega)$.
Moreover, eq. (2) is now valid for $0 \leq t\alpha_i \leq \Omega$:

$$\begin{aligned} F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} nF(t\alpha_i\omega/\Omega)\right] && \text{(by eq. (3))} \\ &= F^{-1}\left[n\sum_{i=1}^{n} F(t\alpha_i\omega/\Omega)\right] \\ &= \frac{\Omega}{\omega}F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i\omega/\Omega)\right] && \text{(by eq. (4))} \\ &= \frac{\Omega}{\omega}tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i\omega/\Omega)\right] \\ & && \text{(by homogeneity applied for } \alpha_i\omega/\Omega \leq \omega) \\ &= tF^{-1}\left[n\sum_{i=1}^{n} F(\alpha_i\omega/\Omega)\right] && \text{(by eq. (4))} \\ &= tF^{-1}\left[\sum_{i=1}^{n} nF(\alpha_i\omega/\Omega)\right] \\ &= tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] && \text{(by eq. (3))} \end{aligned}$$

Repeatedly applying the arguments, extends the intervals in which F is defined and in which homogeneity (eq. (2)) is valid by a factor of Ω/ω each time. Thus, we can extend F to $[0, \infty)$ while preserving homogeneity for $0 \leq t \leq 1$.

Part 3.
The function obtained through this extension must therefore coincide with the original F on at least the interval $[0, \Omega)$. Furthermore, applying eq. (4) on the mean-value function $F^{-1}\left[\left(\frac{1}{n}\sum_{i=1}^{n} F(\alpha_i)\right)\right]$ yields:

$$F^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(\alpha_i)\right]. \tag{5}$$

According to (**hardy1952inequalities**), the only monotonic functions satisfying this equation for $0 \leq \alpha < \infty$ are functions of the form $k\alpha^r$ (see also **aczel1966lectures**).

Therefore, $F(\alpha) = k\alpha^r$ must hold on at least the interval $[0, \Omega]$. Now, for any $|\varphi_i(a_i) - \varphi_i(b_i)|$, we can find $t > 0$ small enough such that $t|\varphi_i(a_i) - \varphi_i(b_i)| < \omega, i = 1, \ldots, n$. Applying monotonicity yields (we can assume $t \leq 1$):

$$\begin{aligned}
\delta(a, b) &= F^{-1}\left(\sum_{i=1}^{n} F\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t}F^{-1}\left(\sum_{i=1}^{n} F\left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t}\left(\sum_{i=1}^{n} \left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)^r\right)^{1/r} && (\text{ since } t|\varphi_i(\alpha_i) - \varphi_i(\beta_i)| \leq \omega) \\
&= \left(\sum_{i=1}^{n} |\varphi_i(a_i) - \varphi_i(b_i)|^r\right)^{1/r}
\end{aligned}$$

Thus, δ is an l^p metric and, by Theorem Todo: ref add-diff-prox-struct-repr , $F = k \cdot \alpha^r$ and hence $F_i = k(t_i\alpha)^r$ or $F_i(\alpha) = k\alpha^r$ if we incorporate the t_i into the φ_i. Finally, the restriction $1 \leq r < \infty$ follows from the triangle inequality or, more precisely, from Minkowski's inequality.

□

6 Discussion

References

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